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TDK (6762) | Japan

Passive Components Factory Tour

Senior Executive Officer Shigeki Sato explained during the factory tour that mass production of ultra-low-profile thin-film inductors embedded in the core layers of semiconductor package substrates for AI servers will begin at the Kofu plant within 1-2 years.

Key Takeaways

- TDK is mass-producing and shipping thin-film inductors for a variety of applications by leveraging thin-film coil fabrication technologies originally developed for HDD thin-film heads.
- Inductors embedded in semiconductor package substrates for GPUs and other devices require high current density, ultra-compact size, and a low-profile form factor, giving TDK's thin-film inductors a competitive advantage.
- TDK has ~50% share of long L-type aluminum electrolytic capacitors used in AI server power supplies. Its proprietary sintered aluminum foil technology provides a competitive advantage in capacitance density.

On Sep 11, we joined a factory tour and passive components business briefing covering TDK's Honjo Nishi Site in Akita Prefecture (which manufactures MLCCs), and Sakata Plant (which produces wound inductors and EMC filters) and Tsuruoka Higashi Plant (which manufactures thin-film filters) in Yamagata Prefecture.

Revenue from passive components for AI servers and data centers expanding at a 30% CAGR: We are particularly focused on the following passive component products for AI servers and data centers: (1) ultra-thin thin-film inductors embedded in semiconductor package substrates, (2) thin-film inductors for optical transceivers, (3) snap-in type long L-type aluminum electrolytic capacitors, and (4) high-voltage MLCCs. Shigeki Sato, Director and Senior Executive Officer responsible for the Electronic Components Business Group, explained that revenue from TDK's passive components for AI servers and data centers is expected to expand at roughly a 30% CAGR over the next 3 years.

¥40bn of capex for thin-film inductors by F3/28: On Sep 11, the *Nikkei Shimbun* reported that TDK will invest ¥35bn in the Kofu plant and ¥5bn in the Tsuruoka Higashi Plant to expand thin-film inductor production (TDK has not confirmed). During the factory tour, Senior Executive Officer Sato explained mass production of ultra-low-profile thin-film inductors embedded in core layers of semiconductor package substrates for AI servers will begin at the Kofu plant within 1-2 years. These ultra-low-profile inductors embedded within package substrates are customized products whose specifications depend on GPU/CPU package substrate designs, and ASPs appear to be significantly higher than those of standard thin-film inductors.

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TDK (6762.T, 6762 JT)	
Electronic Components Japan	
Stock Rating	Overweight
Industry View	In-Line
Price target	¥4,900
Shr price, close (Sep 11, 2026)	¥2,809
Mkt cap, curr, basic (bn)	¥5,331.0
Avg daily trading value (bn)	¥31.6

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The Tsuruoka Higashi Plant will increase production of thin-film inductors used in power circuits for optical modules in AI servers. TDK's thin-film inductors apply thin-film coil fabrication techniques originally developed for HDD thin-film heads and are being mass-produced for a wide range of applications. Because inductors embedded in semiconductor package substrates for GPUs and similar devices require high current density, ultra-compact dimensions, and extremely low profiles (100–300µm), we expect expanding business opportunities for TDK's thin-film inductors.

Capacitors control voltage, inductors control current: Both capacitors and inductors play critical roles in electronic circuits by temporarily storing energy and removing noise. Capacitors act like miniature rechargeable batteries, releasing stored charge to prevent voltage drops and maintain stable voltage levels. Inductors release energy stored in magnetic fields to suppress current fluctuations and maintain stable current flow. MLCCs, of which approximately 25,000 are installed in an AI server, primarily serve 3 functions: (1) stabilizing voltage supplied to GPUs, (2) providing instantaneous current to GPUs, and (3) filtering high-frequency noise. Semiconductor miniaturization improves circuit operating speeds while reducing operating voltages. However, as voltages decline, even small voltage fluctuations can cause semiconductor malfunction, making stable power delivery increasingly important. With every new GPU generation, rising computational performance and power consumption increase capacitance requirements, while board area and available mounting space remain constrained. As a result, we expect continued miniaturization and increased capacitance of MLCCs. Inductor demand in AI server and data center applications has been expanding, although the pace of growth has been slower than that of MLCCs through the Blackwell generation. We expect demand for high-current power inductors to increase substantially from the Rubin generation onward. We forecast rapid growth in demand for high-current power inductors used in vertical power delivery modules in Rubin, and for ultra-thin high-current power inductors embedded in GPU package substrates in Rubin Ultra. Because power inductors embedded in semiconductor package substrates for Rubin Ultra and ASICs require low-profile designs that are difficult to achieve with wound inductors, we expect expanding business opportunities for TDK's thin-film inductors.

TDK holds ~50% share of long L-type aluminum electrolytic capacitors for AI server power supplies: We estimate that TDK's aluminum electrolytic capacitor revenue in F3/26 was only slightly more than 40% of Nippon Chemi-Con's ¥120bn. Nevertheless, TDK explained that it holds an approximately 50% global market share in long L-type aluminum electrolytic capacitors used in AI servers, a market experiencing rapid demand growth. While large numbers of MLCCs are installed near AI server GPUs, we expect demand for high-capacitance aluminum electrolytic capacitors to continue expanding in power modules and UPS systems. TDK stated that its high-capacitance aluminum electrolytic capacitors benefit from proprietary sintered aluminum foil technology, which maximizes aluminum foil surface area through a sintering process using aluminum particles, giving it an advantage in capacitance density over competitors including Rubycon (unlisted), Nippon Chemi-Con (6997.T), and Nichicon (6996.T). TDK also indicated that it has a clear path to increasing capacitance density by a further 20%. The company plans to expand production capacity for long L-type aluminum electrolytic capacitors used in AI

servers to 2x F3/26 levels in F3/27, 4.5x in F3/28, and 7x in F3/29.

Valuation Methodology and Risks

TDK (6762.T)

Derived from the base case. DCF assumptions: risk-free rate 2.6%, equity beta 1.31, risk premium 3.2%, yielding WACC 6.5%. 0% perpetual growth rate from F3/36.

Risks to Upside

- Sharp growth in sales of CATL tech-based rechargeable batteries in devices beyond mobile
- Sharp growth in sales of elec components & rechargeable batteries for new applications like smart glass

Risks to Downside

- TDK's passive components are used in all types of elec equipment, so demand shifts with global economic growth, and resulting supply/demand changes can sway the stock
- ¥1/\$ change affects annual OP by ¥2bn

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(as of August 31, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1555	42%	450	48%	29%	767	44%
Equal-weight/Hold	1577	43%	395	42%	25%	770	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	559	15%	93	10%	17%	216	12%
Total	3,694		939			1754	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Stock Price, Price Target and Rating History (See Rating Definitions)

TDK (6762.T) - As of 09/12/26 GMT in JPY
Industry : Electronic Components



Stock Rating History: 9/1/21 : E/I; 8/2/22 : O/I; 9/8/23 : O/A; 4/10/24 : O/I

Price Target History: 7/29/21 : 913.33; 11/2/21 : 970; 1/17/22 : 1000; 6/30/22 : 900; 8/2/22 : 1100; 9/5/22 : 1200; 11/1/22 : 1260; 1/17/23 : 1160; 1/31/23 : 1100; 3/22/23 : 1140; 6/7/23 : 1280; 9/8/23 : 1260; 11/1/23 : 1380; 11/30/23 : 1560; 1/17/24 : 1600; 2/1/24 : 1760; 6/12/24 : 1940; 7/30/24 : 2220; 9/20/24 : 2160; 11/1/24 : 2250; 1/22/25 : 2350; 1/31/25 : 2300; 6/13/25 : 2100; 9/17/25 : 2600; 12/1/25 : 3100; 1/19/26 : 2700; 3/4/26 : 3200; 6/17/26 : 4900

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
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INDUSTRY COVERAGE: Electronic Components

COMPANY (TICKER)	RATING (AS OF)	PRICE* (09/11/2026)
Shoji Sato		
ALPS ALPINE (6770.T)	O (03/17/2026)	¥2,216
Hamamatsu Photonics (6965.T)	E (06/17/2026)	¥2,193
Ibiden (4062.T)	O (08/04/2026)	¥19,560
Kyocera (6971.T)	E (06/25/2020)	¥3,551
Mabuchi Motor (6592.T)	E (11/03/2022)	¥1,659
Minebea Mitsumi (6479.T)	E (10/23/2025)	¥3,391
Murata Manufacturing (6981.T)	O (11/26/2025)	¥7,352
Nidec (6594.T)	NR (09/05/2025)	¥2,691
Niterra (5334.T)	O (01/17/2024)	¥7,205
Taiyo Yuden (6976.T)	U (06/17/2026)	¥9,209
TDK (6762.T)	O (08/02/2022)	¥2,809
Sota Harashima		
CMK (6958.T)	E (02/28/2025)	¥719
Daishinku (6962.T)	U (06/17/2026)	¥757
Hirose Electric (6806.T)	O (07/10/2024)	¥24,065
IRISO Electronics (6908.T)	E (08/02/2022)	¥2,750
Japan Aviation Electronics Industry (6807.T)	E (01/17/2024)	¥2,488
KOA (6999.T)	E (06/17/2026)	¥2,107
Meiko Electronics (6787.T)	E (04/03/2026)	¥19,020
Nichicon (6996.T)	E (11/10/2021)	¥2,528
Nihon Dempa Kogyo (6779.T)	E (03/07/2024)	¥2,844
Nippon Chemi-Con (6997.T)	U (09/20/2024)	¥2,674

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.